Last weeks:

- Vertex coloring and edge coloring
- Chromatic numbers
- Critical graphs
- Factorization

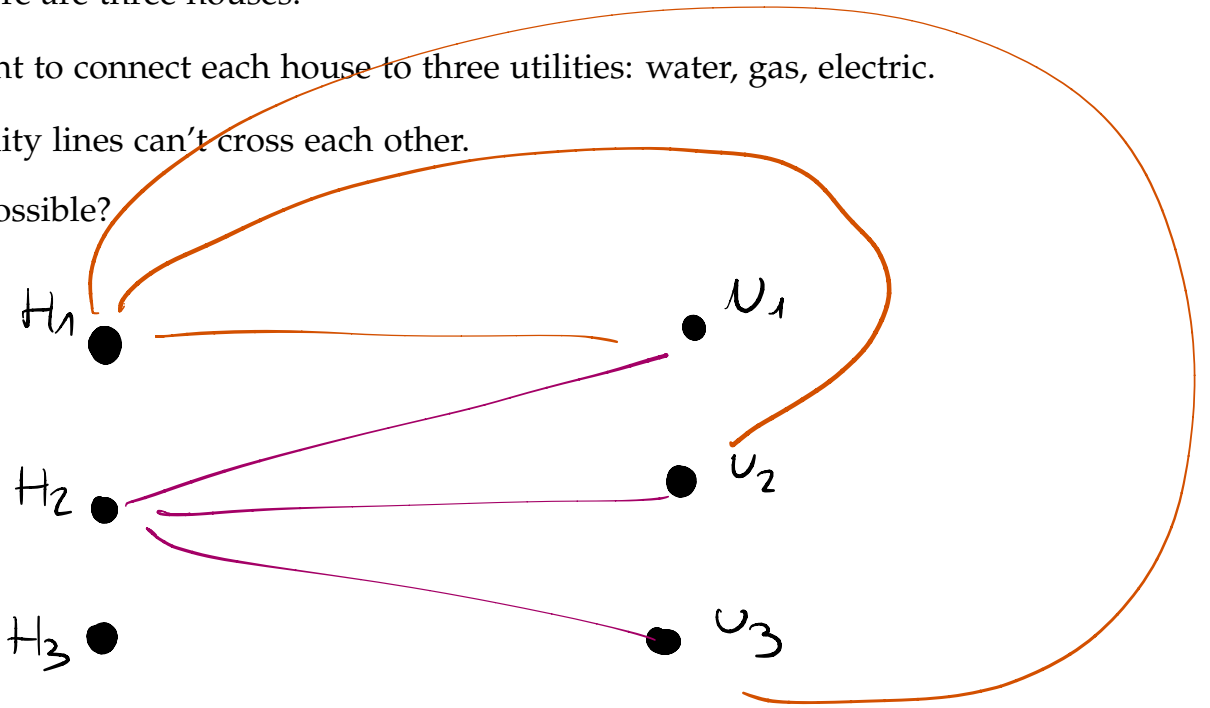
This week:

- Planar graphs
- Planarity and chromatic numbers

Motivation:

- There are three houses.
- Want to connect each house to three utilities: water, gas, electric.
- Utility lines can't cross each other.

Is that possible?



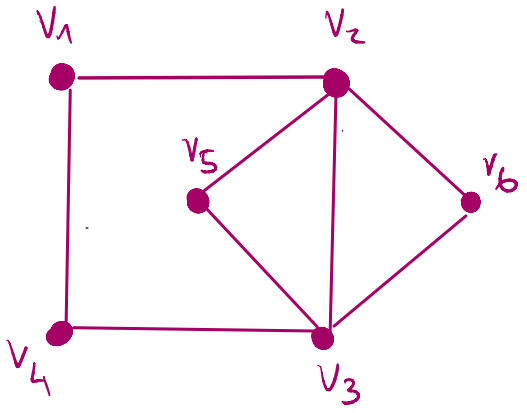
Definition. • *A graph is called **planar** if it can be drawn in the plane so that no two edges cross each other.*

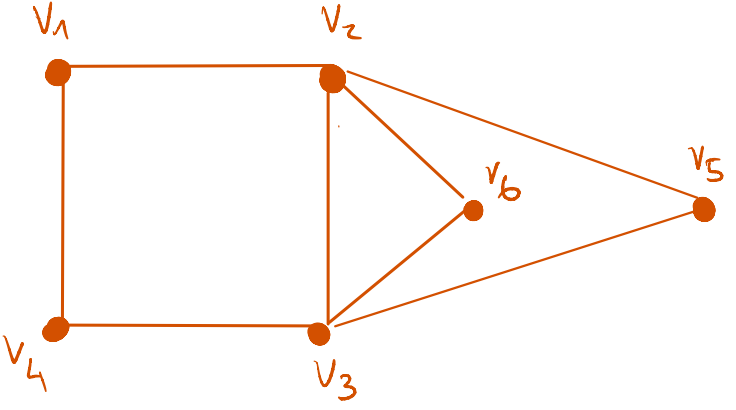
• *A **plane** graph is a graph that is drawn in the plane so that no two edges cross each other.*

Note: A plane graph is just a valid drawing of a planar graph.

Example (A planar graph can have more than one drawing.).

same graph
different(?) drawings



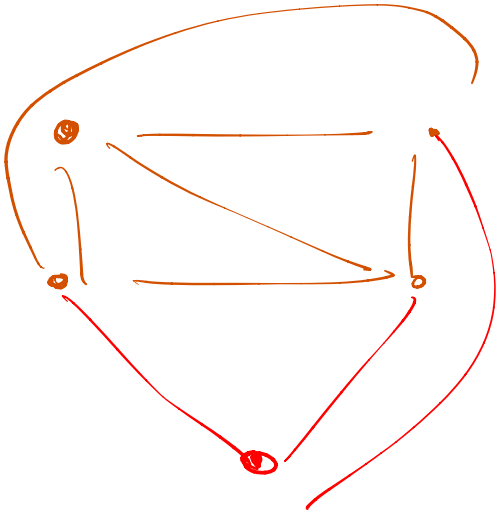


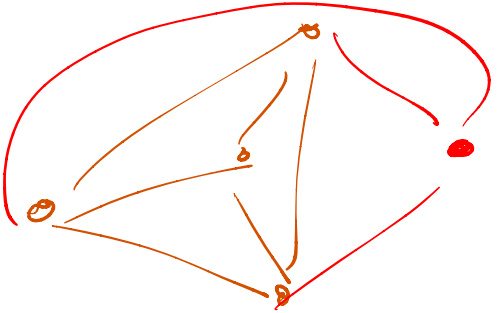
Note: The motivating problem translates to:

Is $K_{3,3}$ Planar?

1) Thm: G is planar $\Rightarrow$ There is some valid drawing using lines

K_4:





Facts and Definitions Let G be a plane graph. Then

(1) G breaks the plane into a collection of connected regions. These regions are called the **faces of** G.

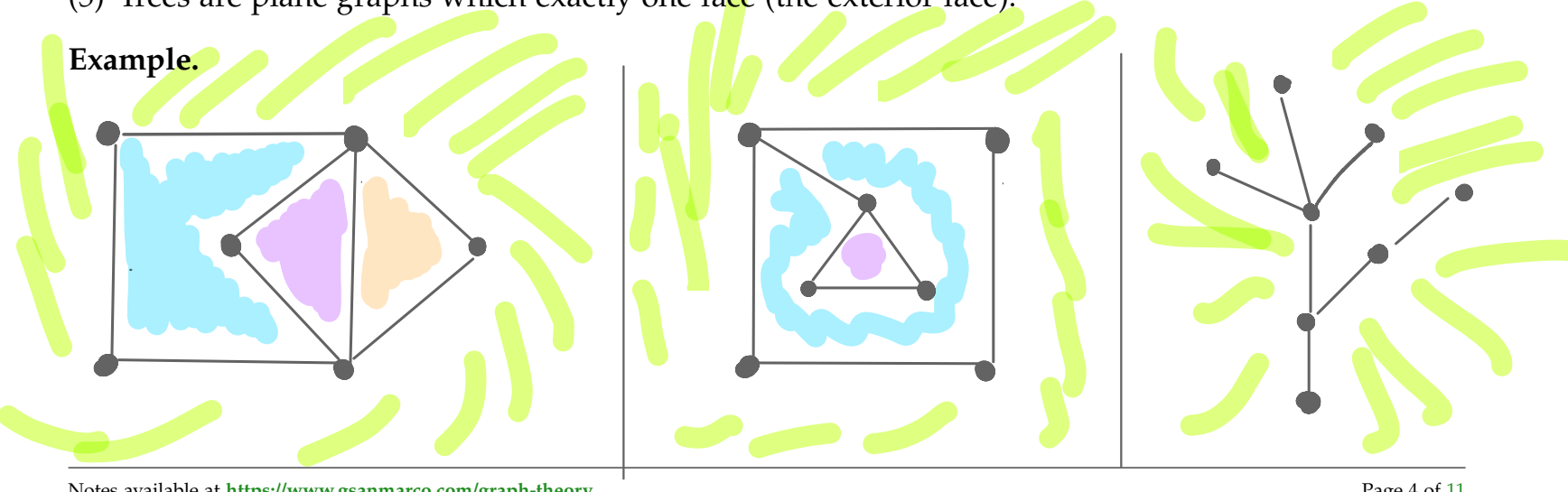
(2) The set of all faces is denoted $F(G)$.

(3) There is exactly one unbounded face, which we refer to as the **exterior face**.

(4) All other faces are called **interior faces.**

(5) Trees are plane graphs which exactly one face (the exterior face).

Example.



More facts and Definitions Let G be a plane graph. Then

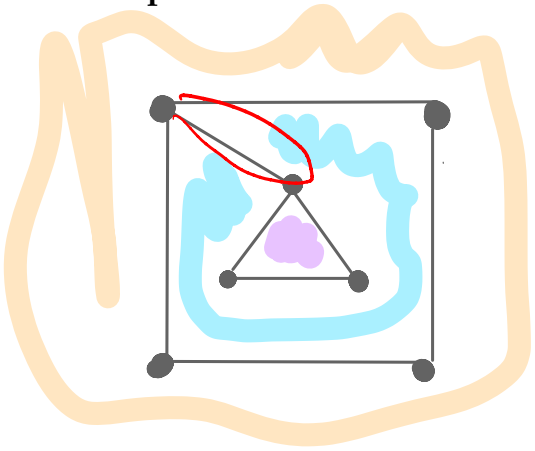
(6) A vertex/edge is **incident** to a face if that vertex/edge lives on the boundary of that face.

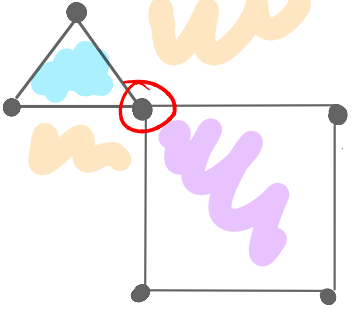
(7) A face could be incident to many vertices, and a vertex could be incident to many faces.

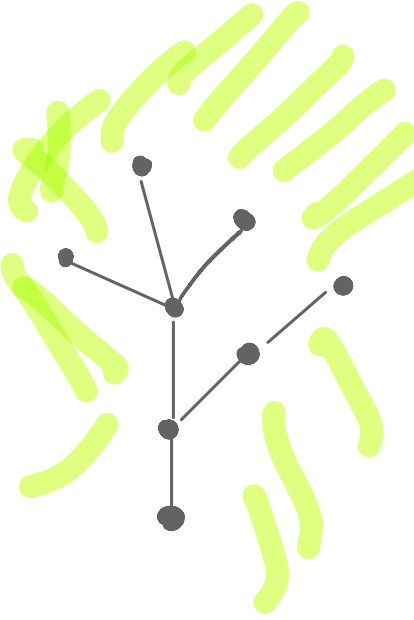
(8) An edge can be incident to at most two faces.

(9) An edge e is incident to exactly one face of G if and only if e is a bridge.

Example.







More facts and Definitions Let G be a plane graph. Then

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(7) A face could be incident to many vertices, and a vertex could be incident to many faces.

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Example

If e is incident to just one face, upon deleting e, then the face will encapsulate some vertices, and vertices outside of it can't connect to vertices inside.

If e is incident to two faces, one is internal, walking around the boundary of that face yields a cycle containing e.

Theorem (Euler's formula, 9.1). *If* G *is a connected plane graph, then*

$$|V(G)| + |F(G)| - |E(G)| = 2.$$

Proof: Induct on $m = |E(G)|$. Put $n = |V(G)|$.

Since G is connected, $m \geq n-1$

Base case: $m = n-1$, G is a tree (connected), Thus it has only one face, so $|F(G)| = 1$.

$$|V| + |F| - |E| = n + 1 - (n-1) = 2 \checkmark$$

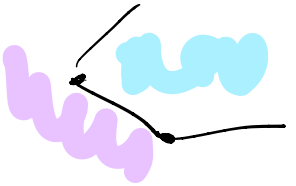
Assume $m \geq n$. Since G is connected, There is a cycle C in G. Let e edge in C, so e NOT a bridge

Let $H = G - e$, so H is plane, H connected, $|V(H)| = n$,

$|E(H)| = m-1$

Claim: $|F(H)| = |F(G)| - 1$

Since e is not a bridge, it is incident to two faces of G but upon deleting e, these two faces merge to a unique face of H.



Every other face of G is untouched ☐ claim.

By induction

$$|V(H)| + |F(H)| - |E(H)| = 2$$

$$n + |F(G)| - 1 - (m-1) = 2$$

$$n + |F(G)| - m = 2$$

$$|V(G)| + |F(G)| - |E(G)| = 2 \quad \square$$

Theorem (Euler's formula). *If* G *is any plane graph, then*

$$|V(G)| + |F(G)| - |E(G)| = 1 + \#\{\text{components of } G\}.$$

Idea: Base case = forest

Corollary. *If* G *is a planar graph, then any two drawings of* G *on the plane have the same number of faces.*

Proof: If we take a drawing

$$|F(G)| = \underbrace{1 + \#\text{components} + |E(G)| - |V(G)|}_{\text{Independent of drawing.}}$$

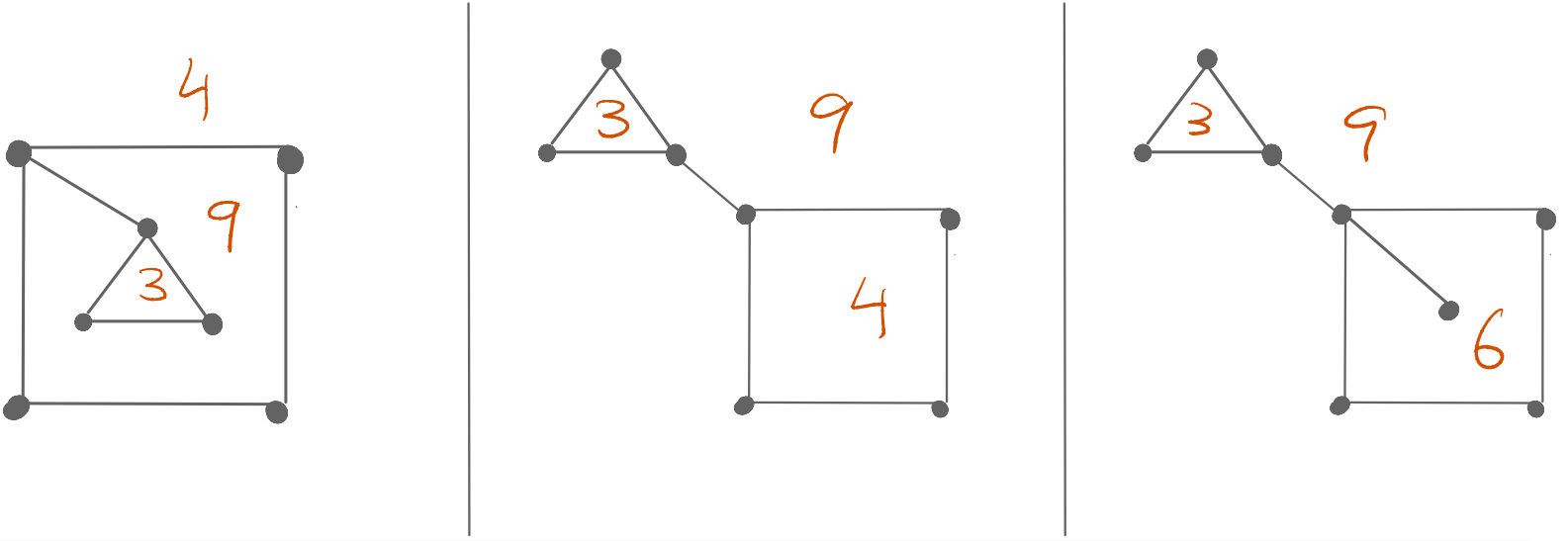
Definition. *Let* G *be a plane graph and let* $f \in F(G)$*. The* ***length of*** f *is*

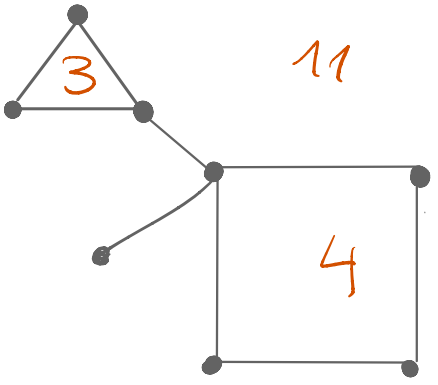
$$len(f) = \#\{\textit{edges incident to } f\} + \#\{\textit{bridges incident to } f\}.$$

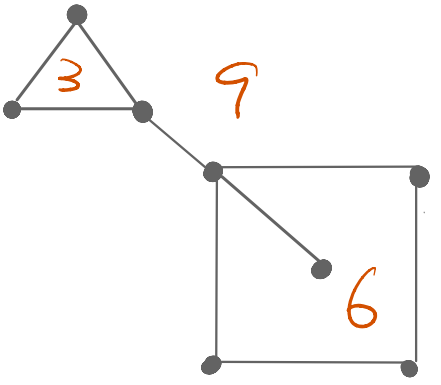
Intuition: Walk around the boundary of the face, counting each edge you encounter.
Bridges are counted twice, because you need to cross it twice to walk around the boundary.

Warning: This depends on the drawing, not just the graph.

Example.




3
11
4


3
9
6

Headshaking Lemma. If G is a plane graph, then

$$\sum_{f \in F(G)} \text{len}(f) = 2|E(G)|.$$

Theorem (9.2). *Let G be a connected planar graph on $n \geqslant 3$ vertices. Then*

$$|E(G)| \leqslant 3n - 6.$$

Proof: Fix a drawing for G

Note: G can't have faces of length 1, 2.

- Length 1:
- Length 2: Boundary → Two non-bridges ✗
  → one bridge

not possible $G \neq K_2$

So $\text{len}(f) \geqslant 3$ $\forall f$

Then By Headshaking

$$2|E(G)| = \sum_{f \in F(G)} \text{len}(f) \geq \sum_{f \in F(G)} 3 \geq 3|F(G)| \Rightarrow$$

$$\Rightarrow |F(G)| \leq \frac{2}{3}|E(G)|$$

By Euler's

$$2 = |V| + |F| - |E| \leq n + \frac{2}{3}|E| - |E| = n - \frac{1}{3}|E|$$

$$\frac{1}{3}|E| \leq n - 2 \Rightarrow |E| \leq 3n - 6.$$

□

Corollary (9.4). *The clique* K_5 *is not a planar graph.*

Proof: K_5 connected, $|V(K_5)| = n = 5$, $|E(K_5)| = \binom{5}{2} = 10$

$3n - 6 = 15 - 6 = 9 < |E(K_5)|$

Thus K_5 NOT planar. □

Question: Is $K_{3,3}$ planar? ??

$|V(K_{3,3})| = 3 + 3 = 6$, $|E(K_{3,3})| = 3 \cdot 3 = 9$

$K_{3,3}$ is connected

$3n - 6 = 3|V(K_{n,n})| - 6 = 12 > |E(K_{3,3})|$

Consistent with $3n - 6 \geq |E(G)|$

Theorem (9.2). *Let* G *be a connected planar graph on* $n \geqslant 3$ *vertices. If* G *is triangle-free, then*

$$|E(G)| \leqslant 2n - 4.$$

$3n-6$

Proof: As before len 1, 2 is not possible

Now, $\text{len}(f) = 3$ is not possible.

$\text{len}(f) = 3$ gives a Triangle.

Thus $\text{len}(f) \geq 4$ $\forall f$. Thus Handshaking

$2|E| \geq 4|F| \Rightarrow |F| \leq \frac{1}{2}|E|$

Euler's: $2 = |V| + |F| - |E| \leq n + \frac{1}{2}|E| - |E| = n - \frac{1}{2}|E|$

$\frac{1}{2}|E| \leq n-2 \Rightarrow |E| \leq 2n-4$ ☒

Corollary (9.5). *The complete bipartite graph* $K_{3,3}$ *is not a planar graph.*

Proof: Connected ✓ $n = 6$, $|E| = 9$, Triangle free.

$$2n - 4 = 8 \leq 9 = |E|$$

Thus not planar ☐